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BINARY SEARCH TREES (BST) & TRAVERSALS

A Binary Search Tree is a hierarchical node-based binary tree data structure with the BST ordering invariant:
- The left subtree of a node contains only keys less than the node key.
- The right subtree of a node contains only keys greater than the node key.
- Both left and right subtrees must also be valid binary search trees.

Traversals:
- Inorder (Left, Root, Right): Traverses the keys in strictly ascending sorted order.
- Preorder (Root, Left, Right): Often used for cloning and serialization.
- Postorder (Left, Right, Root): Useful for tree deletion and bottom-up evaluation.

Time Complexity:
- Balanced Tree (AVL / Red-Black): Search, insertion, and deletion operate in O(log N) time.
- Degenerate Tree (sorted input): Tree degenerates into a singly linked list of height N, degrading search to O(N).